

Diskrete Mathematik

Woche 4 - HG G 26.5

<https://github.com/tschai-yim/eth-resources>



Kittitschai Yimvuthikul

ETH Student TA

Heutige Agenda

1. Teil (45 min)

- 10 min: Fragen zu alten Übungen
- 25 min: Theorie-Übersicht mit Beispielen
- 10 min: Übungen

10 min Pause

2. Teil (45 min): Übungen

Korrektur Tipps

- **Nutzt KI** as *Hilfe*: [00_General/prompts/06d Korrektur.md](#)
 - Vorsicht: Nicht immer genügend streng!
- **Jeder Schritt braucht eine Begründung:**
 - Wenn Lemma verwendet → **immer “from lecture”** angeben
 - Wenn kein Lemma, aber offensichtlich → mit allgemeiner Definition begründen
 - z.B. in $\mathbb{N}$: $y = x + 1 \Rightarrow x < y$ by prop of $\mathbb{N}$
 - Wenn extrem extrem offensichtlich → “holds trivially”
 - z.B. am Ende: thus $2 \neq 1$ trivially
- Wenn immer noch unsicher → mich oder andere TAs fragen!

Fragen zu alten Übungen?

Theorie-Übersicht

Kapitel 2.4 - 2.6

Predicate Logic (recap)

$$\exists x(x > 1 \wedge \forall y \forall z((y \cdot z = x) \rightarrow (y = 1 \vee z = 1)))$$

- **Motivation:** Extend propositional logic to cover more things.
- Same two components:
 - **Syntax:** How you write formulas
 - **Semantics:** What formulas mean

Predicate Logic: New Syntax (recap)

- **Universe U :** Non-empty set of objects to reason about (e.g., $\mathbb{N}$, $\mathbb{R}$).
- **Constants:** Specific, fixed elements from U (e.g., 0 , 1 , π).
- **Predicates $P(x)$:** Functions mapping elements from U to a truth value, $\{0, 1\}$.
 - Example: $\text{less}(x, y)$ returns 1 if $x < y$.
- **Functions $f(x)$:** Map elements from U to another element in U .
 - Example: $\text{add}(x, y)$ on $\mathbb{N}$ returns $x + y$.
- **Variables:** Placeholders for elements from U (e.g., x, y, z).
- **Quantifiers:** $\forall x(\dots)$ and $\exists x(\dots)$

Def 2.10. A k -ary predicate P on U is a function $U^k \rightarrow \{0, 1\}$.

Predicate Logic: New Semantics (recap)

Three layers:

1. **Statements:** English, $\Rightarrow$, $\Leftrightarrow$
2. **Logic** $\{0, 1\}$: Predicate Logic, $\rightarrow$, $\leftrightarrow$
3. **Universe** U : specific interpretation

Example:

$$\forall x(E(x, c) \rightarrow O(m(x), c))$$

1. Let $U = \mathbb{Z}$, $c = 0$, $m(x) = x + 1$, $E(a, b) \Leftrightarrow a$ even, and $O(a, b) \Leftrightarrow a$ not even.
2. Let $U = \mathbb{N}$, $c = 1$, $m(x) = 0$, $E(a, b) \Leftrightarrow a \geq b$, and $O(a, b) \Leftrightarrow a \geq b$.

$\Rightarrow$ satisfiable but not a tautology



Proof Patterns

- Templates to proof things
- **Do not memorize by heart!** Intuition comes naturally with practice.
- First major application of propositional logic

Example indirect proof / contrapositive:

Lem 2.6. $\neg B \rightarrow \neg A \models A \rightarrow B$

Proofs for Implications ($S \Rightarrow T$)

- **Direct Proof** (Def. 2.13): Logically derive T .
- **Indirect Proof / Contrapositive** (Def. 2.14):
 1. **Assume** conclusion T is false ($\neg T$).
 2. Derive that premise S is false ($\neg S$).
- **Composition of Implications / Transitivity** (Def. 2.12):
 - Chain implications: $S \Rightarrow T \Rightarrow U$

Lem 2.5. $(A \rightarrow B) \wedge (B \rightarrow C) \models A \rightarrow C$

Proofs for General Statements (S)

- **Modus Ponens** (Def. 2.15): Prove helper R , then prove $R \Rightarrow S$ to conclude S .
- **Case Distinction** (Def. 2.16): Show $R_1 \vee \dots \vee R_k$, then prove $R_i \Rightarrow S$ for each i .
 - **Example:** For any real number x , $|x| \geq x$
- **Contradiction** (Def. 2.17): Assume $\neg S$, derive a known falsehood T to force S .
 - **Example:** There is no largest natural number.

Proofs involving Quantifiers

- **Existence Proofs** (Def. 2.18): Prove $\exists x P(x)$
 - **Constructive:** Find concrete a where $P(a)$ holds.
 - **Non-constructive:** Prove an example must exist without identifying it.
 - **Example:** See pigeonhole example below.
- **Pigeonhole Principle** (Thm. 2.10)
 - n objects into $k < n$ bins guarantees one bin has $\geq \lceil \frac{n}{k} \rceil$ objects.
 - **Example:** In a room of 367 people, two must share a birthday.
- **Proof by Counterexample** (Def. 2.19): Disprove $\forall x P(x)$
 - **Constructive existence proof** for negation $\neg P(a)$.
 - **Example:** $x = \frac{1}{2}$ disproves $\forall x \in \mathbb{R} (x^2 \geq x)$, since $(\frac{1}{2})^2 = \frac{1}{4} < \frac{1}{2}$.

Proof by Induction ($\forall n \in \mathbb{N} : P(n)$)

- **Principle:** Proves $\forall n \in \mathbb{N} : P(n)$
 1. **Basis Step:** Prove the initial case holds (e.g., $P(0)$ is true).
 2. **Induction Hypothesis (IH):** Assume $P(n)$ holds for an arbitrary $n \in \mathbb{N}$.
 3. **Induction Step:** Prove that $P(n) \Rightarrow P(n + 1)$ using the IH.
 4. **Conclusion:** Conclude that $P(n)$ holds for all $n \in \mathbb{N}$ (Thm. 2.11).
- **Analogy (Domino Effect):**
 - Knock down the first domino (basis).
 - Ensure every falling domino knocks down the next (step).
 - Result: all dominos fall (conclusion).

Übungen

<https://github.com/tschai-yim/eth-resources> unter
03_TA/26HS_252-0025-01L_Diskrete_Mathematik/exercises

Offene Fragen & Feedback

- Gibt es noch Fragen zu den heutigen Themen oder den Übungen?
- Habt ihr Feedback zur heutigen Übungsstunde?
 - Anonymes Formular: <https://forms.gle/xnkcjsdj86LK6yMh7>
- Bei Fragen: Schreibt mir eine E-Mail oder auf WhatsApp.

Schöne Woche und bis nächsten Montag!